

Uttam Shrestha

- **Student ID:** s396560
- **Role:** Project Lead
- **Group:** Group 4
- **Unit:** PRT661 Data Science Practice, Assessment 3

Responsibilities

Project Lead for Assessment 3: planned Sprint 3, set direction of analysis for A3 work, created forecast to decision analysis in notebook 06, coordinated deck, speaker plan, review and merge process on GitHub.

Work completed since Assessment 2

Work	Outcome	Evidence
Sprint 3 planning and Jira upkeep	Created and assigned a3 tasks, updated sprint dates, sprint goal and updated statuses	SCRUM-48 , Sprint 3 board
Saved all out of sample forecasts (notebook 05)	113,677 forecasts at 1 h and 24 h, in line with the A2 results	SCRUM-44 , commit b12cec6
Rolling conformal calibration of 80% intervals	Daylight coverage 80.1% (1 h) and 80.0% (24 h) over all folds; latest period 24 h from 70.6% to 79.3%	SCRUM-45 , notebook 06 , Figure 11
Analysis of site telemetry and battery model	Energy balance fixed meter signs; February 2024 Data drift detected; Battery 354 kWh; 82.8% round trip	SCRUM-46 , Figure 12
KPIs, sensitivity and battery dispatch simulation.	Forecast plan saves 22.3% compared to no battery, or \$3,419 more than the best rule without a forecast.	SCRUM-47 , dispatch_results.md , Figures 13 and 14
A3 analysis delivered and team pull requests reviewed	Two commits added to Jira merged and reviewed/approved team pull requests #3 and #4	PR #2 , PR #3 , PR #4
The presentation deck and speaker plan.	16 slides, timed speaker notes; 4:40 minutes per member.	SCRUM-54
Presented slides 10 to 12 and the live demonstration	Site and battery model, dispatch results, demo of notebook 06, GitHub and Jira	SCRUM-58

Links

- **GitHub repository:** github.com/Uttam1618/PRT661-solar-forecasting
- **Pull request #2 (A3 analysis):** github.com/Uttam1618/PRT661-solar-forecasting/pull/2
- **Notebook 06:** [notebooks/06_bess_dispatch.ipynb](#)
- **Dispatch results and KPIs:** [Outputs/dispatch_results.md](#)
- **Jira board:** uttamshrestha1618.atlassian.net, project SCRUM
- **My Jira tasks:** [SCRUM-44](#), [SCRUM-45](#), [SCRUM-46](#), [SCRUM-47](#), [SCRUM-48](#), [SCRUM-54](#), [SCRUM-58](#)

Planned work before Assessment 4

- Replace nightly rule with an optimiser to plan charging and discharging together.
- Implement drift monitoring and scheduled retraining, based on the discovery made in February 2024.
- Use the enhanced forecast of demand (SCRUM-60) in the dispatch simulation and re-run KPIs.
- Lead the assembly of the A4 final professional report.

Lab 4 Evidence and Reflection

Uttam Shrestha, Student ID s396560, Group 4

The screenshot displays the AWS Academy interface for a lab titled "Lab: Storing and Analyzing Data by Using Amazon Redshift". The user is logged in as Uttam Shrestha (s396560). The interface shows a progress bar at 00:16 and a "Submit" button. A table on the right lists tasks and their scores, totaling 30/30. The tasks are: TASK 2 - Cluster created (5/5), TASK 2b - Security group associated (5/5), TASK 3 - Tables created and data loaded (5/5), TASK 4 - Table data queried (5/5), TASK 5 - Queried by using the AWS CLI (5/5), and TASK 7 - IAM user queried data (5/5). The interface also includes a sidebar with navigation options like Dashboard, Courses, Calendar, Inbox, History, and Help.

Task	Score
TASK 2 - Cluster created	5/5
TASK 2b - Security group associated	5/5
TASK 3 - Tables created and data loaded	5/5
TASK 4 - Table data queried	5/5
TASK 5 - Queried by using the AWS CLI	5/5
TASK 7 - IAM user queried data	5/5
Total score	30/30

Lab 4: Storing and Analysing Data by Using Amazon Redshift (AWS Academy, Module 8). Total score 30/30. The date and time is: Wednesday 23 September 2026, 3:44 pm.

What I learned and how I applied it

The lab guided me through a complete analytics store in the cloud: set up an Amazon Redshift cluster, add a security group, create tables, load a large ticket sales dataset from Amazon S3, and query it from the console, from AWS CLI and as a standalone IAM user with limited access. The key takeaway was why columnar storage is appropriate for analysis; when a query accesses only the required columns, such large tables are still cheap and fast to scan.

I used this directly in A3. Each new result generated has been stored in the columnar Parquet (forecasts_hourly, forecasts_calibrated, dispatch_hourly) and notebook 06 reads only the columns it needs, e.g. only the daylight flag from the 30 feature store. The way raw files are loaded into a separate analytical store from object storage is similar to our raw, processed and output zones, and the data is not stored in Git. The IAM step reinforced least privilege, which we practice by not committing credentials. When site data starts expanding or when the A4 dashboard requires shared access, it's logical to continue with the same Parquet tables in a warehouse like Redshift.

Abhisek Tamang

- **Student ID:** s396739
- **Role:** Data Engineer
- **Group:** Group 4
- **Unit:** PRT661 Data Science Practice, Assessment 3

Responsibilities

Data Engineer for Assessment 3: Maintained alignment in the system architecture developed and the documentation of the repository, reviewed team pull requests and presented the group demonstration.

Work completed since Assessment 2

Work	Outcome	Evidence
System Architecture Diagram as built at A3 (Updated)	Diagram 08 includes site telemetry source, A3 layer (calibration, battery model, dispatch simulation) and shows built versus planned parts	SCRUM-49 , diagram 08
README updated for Assessment 3	A3 results section, notebook 06 added in the run order, repository structure and data excluded from the git	SCRUM-50 , README
Changes delivered through a pull request	Diagram and README merged into main	Pull request #4
Reviewed the team pull requests	Checked and approved PR #2 (A3 analysis) and PR #3 (workflow diagrams) before they were merged	PR #2 , PR #3
Presented slides 1 to 5	Title, overview, objectives, updated architecture and the A2 data pipeline recap	SCRUM-56

Links

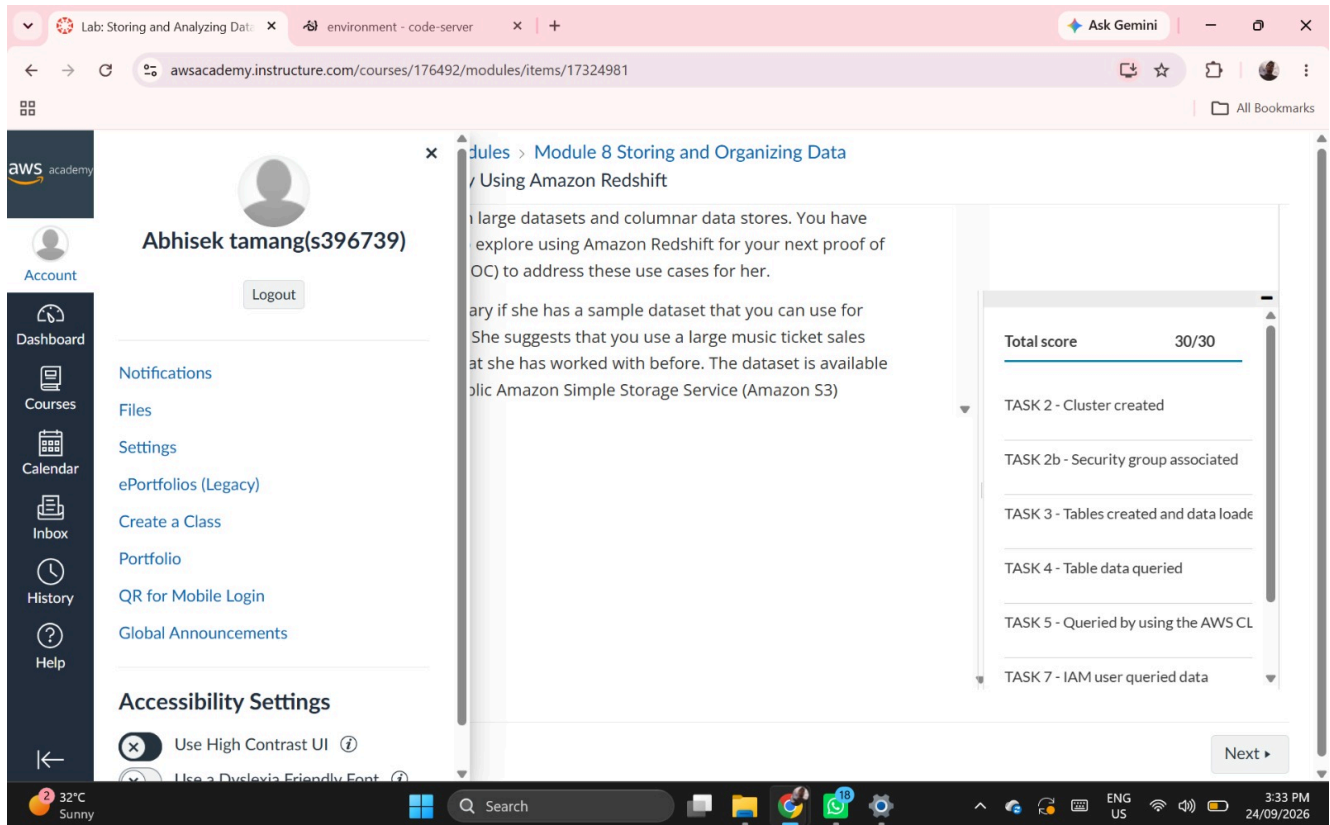
- **GitHub repository:** github.com/Uttam1618/PRT661-solar-forecasting
- **Pull request #4:** github.com/Uttam1618/PRT661-solar-forecasting/pull/4
- **Architecture diagram 08:** [Architecture Diagrams/08-system-architecture-a3.png](#)
- **README:** [README.md](#)
- **Jira board:** uttamshrestha1618.atlassian.net, project SCRUM
- **My Jira tasks:** [SCRUM-49](#), [SCRUM-50](#), [SCRUM-56](#)

Planned work before Assessment 4

- If the scheduled pipeline is available, then create the deployment architecture diagram.
- Maintain architecture diagrams and README through to A4.
- Run the data layer preparation for the A4 dashboard to enable the sharing of the Parquet outputs and query of the data.

Lab 4 Evidence and Reflection

Abhisek Tamang, Student ID s396739, Group 4



Lab 4: Storing and Analyzing Data by Using Amazon Redshift (AWS Academy, Module 8). Total score 30/30. Date and time shown: Thursday 24 September 2026, 3:33 pm.

What I learned and how I applied it

This lab consisted of creating an Amazon Redshift cluster, adding a security group, creating tables and loading a large music ticket sales data set from Amazon S3, and then querying that data set from the console, AWS CLI and as a separate IAM user. It gave me a glimpse of an end-to-end data warehouse, where raw data lives, how it gets into a place where it can be analysed and who can access it.

This influenced my description of our system in A3. The new architecture diagram is split into the same layers that the lab uses – raw extracts as the source, the columnar store in Parquet for analysis and outputs that feed into the decision. In the README I did note that the data remains external to Git, as the lab does with data on S3 rather than in the code. I was also reminded of the IAM and security group that access should only be granted for the individual to the resources they need, which will be important when adding a dashboard in A4.

Sarin Uprety

- **Student ID:** s394012
- **Role:** Data and Feature Engineer
- **Group:** Group 4
- **Unit:** PRT661 Data Science Practice, Assessment 3

Responsibilities

Data and feature engineer for Assessment 3: wrote documentation for data flow through notebooks and how team implements each change, covered team pull requests, and completed group demonstration presentation on workflow, progress and interval calibration.

Work completed since Assessment 2

Work	Outcome	Evidence
Workflow progress chart at A3	Progress by epic with the A3 status of all eight epics	SCRUM-52 , chart 02c
Workflow diagram as built at A3	Data workflow from raw extracts to battery decision and the 7 step team workflow	SCRUM-52 , diagram 04
Changes delivered through a pull request	Both diagrams merged into main	Pull request #3
Reviewed the team pull requests	Checked and approved PR #2 (A3 analysis) and PR #4 (architecture diagram and README) before they were merged	SCRUM-53 , PR #2 , PR #4
Presented slides 6 to 9	Workflow, implementation progress, A2 model recap and the interval calibration	SCRUM-57

Links

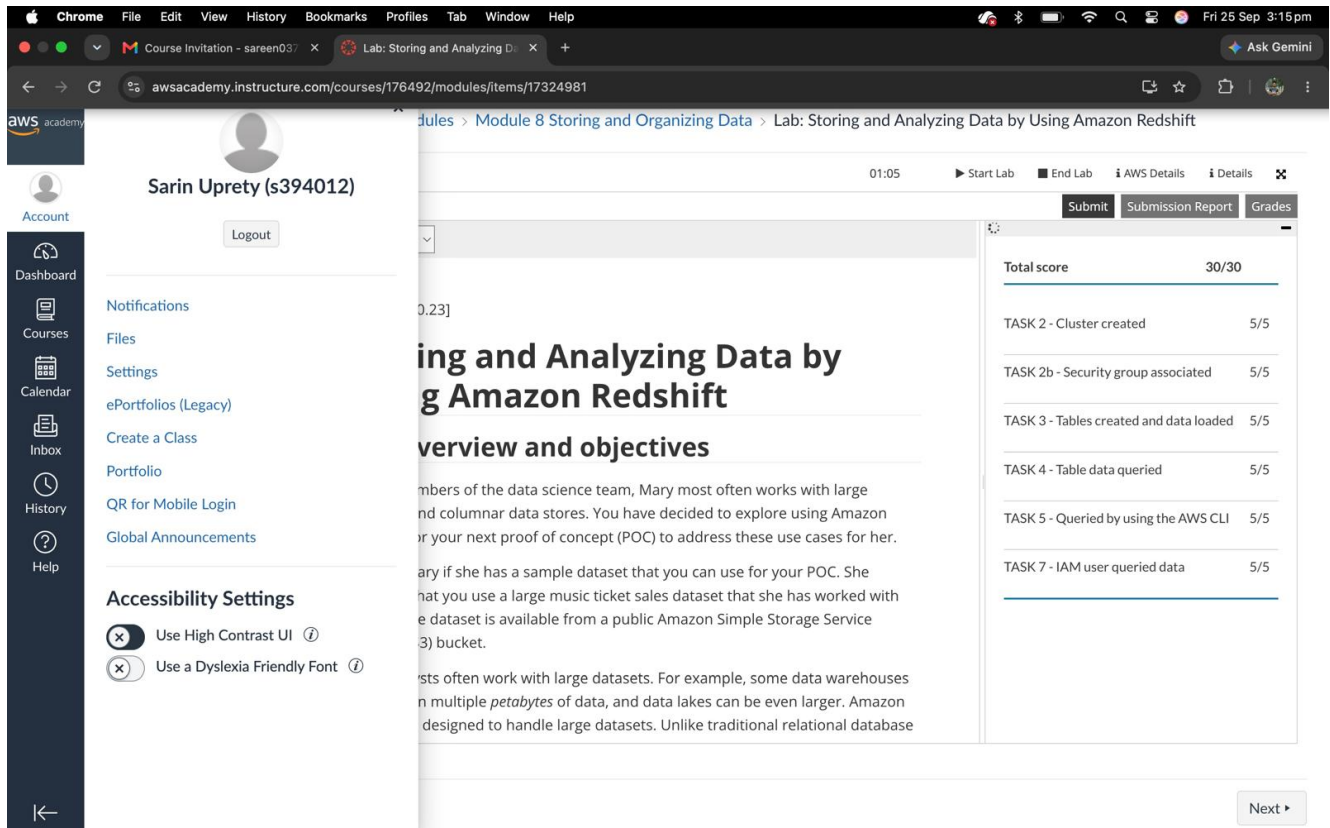
- **GitHub repository:** github.com/Uttam1618/PRT661-solar-forecasting
- **Pull request #3:** github.com/Uttam1618/PRT661-solar-forecasting/pull/3
- **Workflow diagram 04:** [Workflow Diagrams/04-workflow-a3.png](#)
- **Progress chart 02c:** [Workflow Diagrams/02c-workflow-progress-a3.png](#)
- **Jira board:** uttamshrestha1618.atlassian.net, project SCRUM
- **My Jira tasks:** [SCRUM-52](#), [SCRUM-53](#), [SCRUM-57](#), [SCRUM-51](#)

Planned work before Assessment 4

- Run the reproducibility check of notebooks 05 and 06 from a fresh clone and record the results (SCRUM-51).
- Add drift check and retraining step to the data workflow as described in A4.
- Before the broader use of the site telemetry, check the cleaning rules against the 2024 and 2025 site telemetry.

Lab 4 Evidence and Reflection

Sarin Uprety, Student ID s394012, Group 4



The screenshot shows the AWS Academy interface for Lab 4: Storing and Analyzing Data by Using Amazon Redshift. The user, Sarin Uprety (s394012), has completed the lab with a total score of 30/30. The tasks completed are:

Task	Score
TASK 2 - Cluster created	5/5
TASK 2b - Security group associated	5/5
TASK 3 - Tables created and data loaded	5/5
TASK 4 - Table data queried	5/5
TASK 5 - Queried by using the AWS CLI	5/5
TASK 7 - IAM user queried data	5/5

Lab 4: Storing and Analyzing Data by Using Amazon Redshift (AWS Academy, Module 8). Total score 30/30. Date and time shown: Friday 25 September 2026, 3:15 pm.

What I learned and how I applied it

The lab proceeded with the steps to load a large dataset in Amazon Redshift: Create cluster and security group, create tables, copy data from Amazon S3, and query Amazon Redshift data from the console and the AWS CLI and as a limited IAM user. What I found interesting was that the load step relies on the table definition being identical to the data, and that subsequent queries are dependent on this step being correct.

This is what I did when I documented our workflow on a piece of A3 paper. The workflow diagram makes it easy to see where each table is coming from, similar to how the lab traces the data from S3 into the warehouse, by displaying each notebook and the file it reads, and the file it writes. It also ties into with my cleaning work for A2 values: values should be checked and cleaning should be done before loading and aggregation. The same applies to the team workflow side of the diagram: for our code, every change goes through a task, a branch and a reviewed pull request.

Yogesh Basnet

- **Student ID:** s394043
- **Role:** Modelling and Visualisation Lead
- **Group:** Group 4
- **Unit:** PRT661 Data Science Practice, Assessment 3

Responsibilities

Modelling and visualisation lead for Assessment 3: presented technical challenges, limits of result and future work, reviewed team pull requests, created the final team video and pushed final presentation files to GitHub.

Work completed since Assessment 2

Work	Outcome	Evidence
Presented slides 13 to 16	Possible technical problems, what influences the outcome, future work before A4 and A4 summary	SCRUM-59
Reviewed the team pull requests	Checked and approved PR #2 (A3 analysis), PR #3 (workflow diagrams) and PR #4 (architecture diagram and README)	PR #2 , PR #3 , PR #4
Committed the final deck and individual summaries	The presentation (PPTX and PDF) and individual contribution PDFs submitted to the repository via a reviewed pull request.	SCRUM-61
Merged, checked and uploaded the group video	Four clips side by side, reviewed in less than 20 minutes, uploaded and link shared	SCRUM-55
Planned the A4 demand forecast improvement	Task defined as being limited by demand (85% w/ perfect demand, 52% w/ perfect solar)	SCRUM-60

Links

- **GitHub repository:** github.com/Uttam1618/PRT661-solar-forecasting
- **README with the video link:** [README.md](#)
- **Dispatch results used in my slides:** [Outputs/dispatch_results.md](#)
- **Jira board:** uttamshrestha1618.atlassian.net, project SCRUM
- **My Jira tasks:** [SCRUM-55](#), [SCRUM-59](#), [SCRUM-60](#), [SCRUM-61](#)

Planned work before Assessment 4

- Build the improved demand forecast with weekday and weekend patterns and temperature (SCRUM-60).
- Compare it with the current 7 day average and report the new share of the perfect foresight benefit.
- Lead the Visualisation epic towards the A4 dashboard.

Lab 4 Evidence and Reflection

Yogesh Basnet, Student ID s394043, Group 4

The screenshot shows the AWS Academy lab interface for 'Storing and Analyzing Data by Using Amazon Redshift'. The interface is displayed in a Chrome browser window. The top navigation bar includes 'Home', 'Modules', 'Discussions', 'Grades', 'Lucid (Whiteboard)', and 'Notebook'. The main content area is titled 'Storing and Analyzing Data by Using Amazon Redshift' and includes a 'Lab overview and objectives' section. The overview text describes the lab's purpose and the tasks involved. The objectives section lists seven tasks, all of which are completed (5/5). The tasks are: TASK 2 - Cluster created, TASK 2b - Security group associated, TASK 3 - Tables created and data loaded, TASK 4 - Table data queried, TASK 5 - Queried by using the AWS CLI, and TASK 7 - IAM user queried data. The total score is 30/30. The bottom of the interface shows 'Previous' and 'Next' navigation buttons.

Task	Score
TASK 2 - Cluster created	5/5
TASK 2b - Security group associated	5/5
TASK 3 - Tables created and data loaded	5/5
TASK 4 - Table data queried	5/5
TASK 5 - Queried by using the AWS CLI	5/5
TASK 7 - IAM user queried data	5/5

Lab 4: Storing and Analyzing Data by Using Amazon Redshift (AWS Academy, Module 8). Total score 30/30. Date and time shown: Wednesday 30 September 2026, 12:00 pm.

What I learned and how I applied it

The lab demonstrated how I could use a columnar data warehouse to process a large dataset: I created an Amazon Redshift cluster, uploaded a music ticket sales data set to Amazon S3, and then queried the data with the Amazon Redshift Console, the AWS CLI and as an IAM user. A query that requires just a few columns from a large table will still be fast, due to Redshift's columnar structure of storing data. This is relevant to the work I bring to A4. Our A3 analysis found that the demand forecast, not the solar model, limits the battery plan. To improve it requires the ability to add several years of site telemetry, together with temperature and calendar features, and that's the type of wide table where you're just looking for the columns you need — which is the case with our Parquet stores. The lab also gave me some insights into the organisation of the data before modelling, which I think I will have to apply to the creation and comparison of the new demand forecast as I build it.